

Lecture 18

RL + DL Recap

COMP532 – Machine Learning and Bioinspired Optimisation

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Reinforcement Learning

The difference between:

- ▶ Traditional machine learning
- ▶ Reinforcement learning

Reinforcement Learning

- ▶ Multi-armed bandits
- ▶ The exploration/exploitation dilemma
- ▶ Strategies:
 - ▶ Greedy
 - ▶ ϵ -greedy
 - ▶ Random

Reinforcement Learning

- ▶ Value functions
- ▶ State-value function for policy π : $V^\pi(s)$
- ▶ Action-value function for policy π : $Q^\pi(s, a)$
- ▶ Bellman equation
- ▶ The equation is recursive
- ▶ Solving the Bellman Equation:
 - ▶ Dynamic Programming (DP)
 - ▶ Temporal Difference (TD)
 - ▶ Monte Carlo Tree Search (MCTS)

Reinforcement Learning

- ▶ Q-learning
- ▶ Update rule

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left(r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right)$$

- ▶ Exploration vs exploitation
- ▶ Deep Q-learning
- ▶ Use a neural network to approximate $Q(s, a)$
- ▶ Loss function

Exam Example

What is the primary advantage of using Deep Q-Learning over traditional Q-Learning?

- ▶ A) It can only solve problems with discrete action spaces
- ▶ B) It eliminates the need for a replay buffer
- ▶ C) It can effectively handle high-dimensional input spaces
- ▶ D) It guarantees convergence to the global optimum

Correct Answer:

C) It can effectively handle high-dimensional input spaces

Deep Learning

- ▶ Neurons
- ▶ Neural networks:
 - ▶ Nodes
 - ▶ Edges
 - ▶ Weights

Deep Learning

- ▶ Input layer
- ▶ Hidden layers
- ▶ Output layer

Deep neural networks:

- ▶ Input layer
- ▶ Multiple hidden layers
- ▶ Output layer

Deep Learning

- ▶ Activation functions
- ▶ Layers
 - ▶ Convolution layers
 - ▶ Pooling layers
- ▶ Loss functions

Exam Example

What does MSE stand for, and how is it used as a loss function?

Answer:

Mean Squared Error (MSE) is defined as

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- ▶ n = number of samples
- ▶ y_i = true value
- ▶ $\hat{y}_i$ = predicted value

MSE measures the average squared difference between predicted and true values.

Deep Learning

- ▶ Backpropagation
- ▶ Update weights using gradient descent
- ▶ Forward propagation
- ▶ Backward propagation
- ▶ Optimiser

Example Exam Questions

- ▶ Explain the concept of Q-learning.

Deep Reinforcement Learning

Deep Reinforcement Learning combines:

- ▶ Reinforcement Learning
- ▶ Deep Neural Networks

Instead of storing a Q-table:

$$Q(s, a)$$

we learn a neural network approximation:

$$Q_{\theta}(s, a)$$

- ▶ Handles high-dimensional inputs
- ▶ Works with images, raw states, sensors
- ▶ Enables large-scale environments

LLMs as Agents

Modern AI systems often use LLMs as agents.

Agent loop:

1. Observe environment
2. Reason using the LLM
3. Take actions (tool use)
4. Receive feedback

LLM agents can:

- ▶ interact with tools
- ▶ search for information
- ▶ solve multi-step problems